

DIGITAL MIRROR

AI-Powered Multimodal Behavioral Analysis & Simulation Framework

1. Problem Statement

Modern social media platforms are designed to maximise user engagement using algorithm-driven recommendation systems, short-form videos, emotionally triggering content, and addictive visual patterns. Continuous exposure to such content gradually affects human behaviour, emotional stability, cognitive focus, sleep patterns, communication habits, and digital dependency.

Existing digital wellbeing systems mainly focus on screen-time monitoring and application blocking. However, they fail to computationally analyse the deeper psychological and behavioural influence created by algorithmic content exposure.

DIGITAL MIRROR is proposed as a research-oriented AI-powered multimodal behavioural analysis platform that analyses text, audio, image, and video inputs to identify behavioural addiction patterns, emotional instability, attention fragmentation, cognitive fatigue, and algorithmic influence.

Computational Behavioural Indicators:

- Dopamine Trigger Score
- Attention Fragmentation Index
- Sleep Disruption Score
- Emotional Dependency Score
- Digital Behavioral Risk Index (DBRI)

2. Proposed Solution

The proposed solution uses Artificial Intelligence, Behavioural Psychology Modelling, NLP, Computer Vision, and Multimodal Analysis techniques to computationally evaluate the psychological and behavioural impact caused by social media algorithms.

Multimodal Inputs • Text • Audio • Images • Videos	Platform Functions • Sentiment Analysis • Emotional Analysis • Addiction Pattern Detection • Attention Fragmentation • Sleep Disruption Analysis • Behavioural Risk Prediction	Output & Reports • Behavioural Insights • Psychological Risk Reports • Emotional Analytics • Future Behavioural Simulations • Visualization Dashboards
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Calculated Behavioural Metrics:

- Dopamine Trigger Score
- Emotional Dependency Score
- Cognitive Fatigue Score
- Algorithmic Influence Score
- Digital Behavioral Risk Index (DBRI)

3. Workflow

- 1

User uploads text, audio, image, and video inputs
- 2

System preprocesses and cleans the input data
- 3

AI modules analyse emotions, addiction patterns, and behaviour
- 4

Behavioural psychology metrics are calculated
- 5

Behavioural Risk Engine generates risk scores
- 6

Digital Behavioral Risk Index (DBRI) is generated
- 7

Dashboard visualises behavioural insights and analytics

4. Technologies Used

Python	Core development language
Machine Learning	Behavioural pattern modelling
NLP	Natural Language Processing for text analysis
Computer Vision	Visual content & image analysis
Speech Emotion Rec.	Audio emotion detection
OpenCV	Video & image processing
Pandas & NumPy	Data manipulation and computation
Streamlit / Flask	Web application framework
Data Visualization	Dashboards and analytics charts

5. Project Vision

DIGITAL MIRROR is designed to help users understand how algorithm-driven social media platforms gradually influence human psychology, emotional behaviour, cognitive focus, and digital dependency using computational behavioural analysis and AI ethics awareness.

LABEL ANNOTATION LINK

https://drive.google.com/file/d/1w_OGWFiaD-LBXs4lXhT3MOQ8NkPAIPpa/view?usp=sharing

Click the link above to access the label annotation dataset on Google Drive.